# THELAWALA – A VENDOR AGGREGATION PLATFORM

CS23B1062 - Anurag Sharma | CS23I1064 - Parth Pandey | CS23I1026 - Grishmank Parate | ME23B1009 - Shashwat Kumar Chandan

Mentor : Dr. Narendran G.

## The Problem

**Street vending** currently depends on **"random encounters"**, where buying and selling happen by pure luck rather than a plan. This randomness forces customers to **waste time** waiting for arrivals they can't predict, while vendors exhaust themselves walking empty routes with no way to find interested buyers. Without a real-time connection, this luck-based system leads to **missed sales** for the vendor and constant **frustration** for the community.

## Objectives

To establish a **real-time synchronization** system between mobile vendors and semi-urban & rural households, replacing the uncertainty of random encounters with a reliable digital network that provides live location visibility and arrival alerts for both parties.

## Methodolgy & Assumptions

### Methodology: Information Discovery

To build the concept, we **analyzed the informal vending ecosystem** using structured systems-engineering tools:

- **Stakeholder Needs Analysis (SNAC):** Prioritized the **ease of access** for house wives and senior citizens and the **time-sensitivity** of urban professionals.
- **Discovery Matrix:** Cross-referenced system variables to identify **"Reach"** and **"Time"** as the highest-impact nodes (Scores: 16+).
- **ISM Hierarchy:** Determined that the **"Uncertainty of Availability"** is the primary root cause that our concept must address.

### Critical System Assumptions

- **Mobile Connectivity:** It is assumed that vendors possess smartphones with active data plans to facilitate continuous background data transmission.
- **Signal Reliability:** The system assumes that wireless and positioning signals remain stable and accurate within dense urban environments.
- **Behavioral Adoption:** A core assumption is that consumers are willing to transition from traditional "listening and waiting" to a digital-first notification model.
- **Operational Autonomy:** It is assumed that the technology can function for a full operational shift without requiring manual intervention or recharging.

## Concept Design

### Research-Driven Foundation

Analysis using the SNAC, Discovery Matrix, and ISM frameworks revealed that **unpredictable availability**, **zero communication channels**, and a **lack of location awareness** are the root bottlenecks in street vending.

This led to a **real-time platform** where users can **discover** thelawalas currently near them, search for vendors selling specific products (e.g., "panipuri"), and **track** their desired thelawalas on a live map.

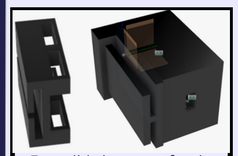
### Seamless Communication Channel

The platform establishes a communication channel where a user can send a direct purchase request for an item. This instantly relays the user's exact location to the thelawala, bridging the gap between vendor routes and active buyer demand.

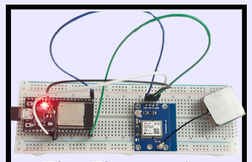
### The Hardware Pivot (AUJAR)

While the system could rely entirely on smartphone hardware, continuously broadcasting **GPS data** causes severe **battery drainage** over 8–12 hour shifts, and a portion of thelawalas might not even own a smartphone. To tackle this we introduced **AUJAR**, a specialized, low-power hardware device mounted directly on the cart. It continuously **acquires GPS data and transmits it** via Bluetooth (BLE) to any paired smartphone nearby (or independently), completely offloading the location-relaying burden and battery drain

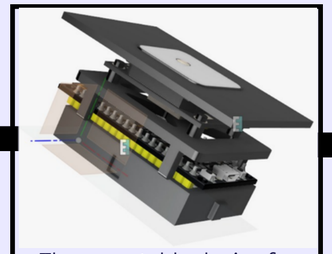
## Prototype



Easy slide in mount for the device

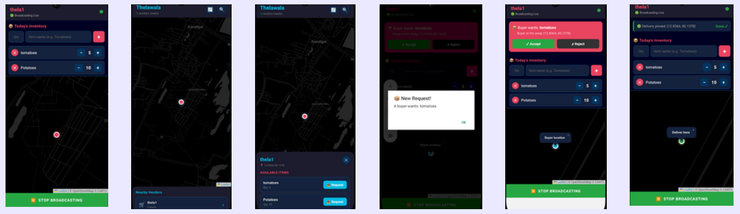


The Electronics



The mountable device for tracking the vendor's cart

## Workflow of the APP



Vendor Home Page | User Vendor Discovery | User Stock Discovery | Request Recieved by the Vendor | Request Acceptance | User Location for delivery

## Bill of Materials

- ESP WROOM 32 MCU Module Version: 1.1
- NEO-6M GPS Module
- 400 pin breadboard
- Jumper Wires

## Observation & Inferences

Buyers and vendors miss transactions due to a lack of real-time location awareness and direct communication.But relying solely on mobile phones for continuous GPS tracking is unviable due to severe battery drain and limited smartphone ownership.

**Project Outcomes -**

- **Application Ecosystem:** Developed functional mobile applications for both Users and Vendors.
- **Bridge Server:** Built a central server architecture designed to connect the apps and route real-time data.
- **Hardware Prototype:** Assembled the AUJAR hardware circuit (ESP32 + GPS module) to serve as a low-power tracking alternative.

## References

- Sneha, M., & Praveen, K., & Myna, K., & Rithika, C. (). GPS BASED WOMEN SECURITY SYSTEM USING ESP32. International Journal of Innovative Research in Technology (IJIRT), 9(11), 221–224.
- Irvan Nauval, R. Arri Widyanto, Agus Setiawan, Nuryanto Nuryanto, Nugroho Agung Prabowo, Tuessi Ari Purnomo; Implementation of GPS for tracking of street vendor. AIP Conf. Proc. 29 November 2023; 2702 (1): 060006